

DUC ANH NGUYEN

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EDUCATION

Simon Fraser University

Bachelor of Science in Computing Science

Burnaby, BC

Sep 2022 – Oct 2027

RESEARCH EXPERIENCE

Selective Co-Presence

2026 – Present

Undergraduate Researcher | Tangent Lab, Simon Fraser University

Burnaby, BC

- Implemented an Apple Vision Pro diminished-reality system with five selective co-presence filters—full-person diminishment, abstraction, silhouette, facial diminishment, and a virtual wall—for a planned study with adults with ADHD.
- Built a stereo person- and face-tracking pipeline using per-eye segmentation, calibrated stereo matching, an OC-SORT adaptation, and temporal filtering to stabilize binocular overlays.
- Adapted MetalSplatter and multi-cone casting to render Gaussian-splat room reconstructions inside tracked person masks while preserving the surrounding passthrough view.

Functionalization via Structure Completion and Motion Rectification

2026

Research Assistant | GrUVi Lab, Simon Fraser University

Burnaby, BC

- Owned end-to-end curation and quality control for FurFun-233: sourced, annotated, and repaired 233 furniture models in Blender for reliable training and evaluation.
- Improved the hinge-snapping heuristic by projecting each 3D hinge region of interest onto a 2D footprint, making connector insertion orientation-robust and extending it from storage furniture to objects such as eyeglasses.
- Validated generated structures and motions against expected behavior and produced the paper's project video, selecting qualitative examples and communicating the end-to-end pipeline.

PROJECTS

VETI: 3D-Guided Visual Effect Tuning

Feb 2026 – Apr 2026

Co-first Author | Accepted SIGGRAPH Asia 2026 Poster | Python, PyTorch, Blender, MoGe-2, SAM 2

- Built a geometry-aware image-editing pipeline that transforms a photograph and object masks into an editable Blender scene using intrinsic decomposition, 2.5D reconstruction, and material controls.
- Implemented unmixing-based soft color segmentation in PyTorch, integrated MoGe-2 and SAM 2, and automated the export of editable layers, masks, meshes, and scene configurations.

Aurora Rendering

Feb 2026 – Apr 2026

Physically Based Rendering Project | Python, Mitsuba 3

- Developed a three-stage volumetric aurora renderer that generates a fluid-simulated footprint, constructs a volume grid, and renders the final scene in Mitsuba 3.
- Implemented Jos Stam's Stable Fluid Solver and extended a path tracer with importance sampling for surface illumination and volumetric rendering.

Object Detection and Instance Segmentation

Feb 2025 – Mar 2025

Computer Vision Project | Python, PyTorch, Detectron2

- Combined object detection and semantic segmentation to perform airplane instance segmentation on high-resolution imagery.
- Adapted a SAHI-inspired tiling method to preserve small visual details during training and improve segmentation accuracy.

Sign Language Translator

Sep 2025 – Nov 2025

Real-Time Machine Learning Project | Python, PyTorch, scikit-learn, MediaPipe

- Developed two real-time webcam classifiers: a convolutional neural network trained on hand-sign images and a K-nearest-neighbors model trained on MediaPipe hand landmarks.

TECHNICAL SKILLS

Languages: Python, C, C++, C#, JavaScript, MATLAB, HTML/CSS

Frameworks/Libraries: PyTorch, Detectron2, scikit-learn, MediaPipe, OpenCV, NumPy, Apple Vision

Tools: Git, Conda, Jupyter, Blender, Mitsuba 3, Apple Vision Pro, MetalSplatter